

What Should We Ask of an Inequality Measure?

Axioms, Proofs, and How Often They Bind in Survey Data

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Abstract

We evaluate eight inequality indices against eleven axiomatic properties, with a proof or an explicit counterexample for every index–property pair, and then ask how often the properties decide anything in real data. On the theory side we show that the Zenga index fails diminishing transfer sensitivity and subgroup consistency, and that the exponential absolute index, like the Generalized Entropy family, satisfies ten of the eleven properties, so counting axioms gives no reason to prefer relative over absolute measures. On the data side we compute every index on 2,588 percentile distributions from the World Bank’s Poverty and Inequality Platform. Across 1,955 consecutive comparable survey pairs in 143 countries, Lorenz curves cross in 62% of pairs; when they cross, relative indices that all satisfy the transfer principle agree on whether inequality rose in only 51% of them, but most of this disagreement concerns small changes. Lorenz dominance settles the comparison for every relative index that satisfies the transfer principle, but not for absolute indices: among pairs with a change of at least one Gini point and dominant Lorenz curves, where all such relative indices must agree, the absolute Gini moves the other way in 20% of cases.

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1 Introduction

How should we measure inequality? The question is as old as welfare economics itself, yet it remains surprisingly unsettled in practice. Applied researchers choose among many indices: the Gini coefficient, the Theil index, the Atkinson index, the variance of logarithms, percentile ratios, and many others. Each appears in thousands of published papers, yet the criteria for choosing among them are rarely made explicit. A researcher studying wage inequality might reach for the variance of logarithms out of habit, unaware that this measure can *increase* after a progressive transfer from a richer individual to a poorer one—a property so counterintuitive that it undermines the very purpose of inequality measurement. Another researcher might favor the Gini coefficient for its elegant geometric interpretation, not realizing that it cannot be cleanly decomposed into within-group and between-group components when subgroups overlap in income.

We argue that the choice of inequality measure should be guided by a clear understanding of axiomatic properties—intuitive desiderata that any “reasonable” measure ought to satisfy. The axiomatic approach to inequality measurement has a distinguished pedigree, originating with [Dalton \(1920\)](#) and [Pigou \(1912\)](#) and formalized in the seminal contributions of [Atkinson \(1970\)](#), [Sen \(1973\)](#), [Kolm \(1976a,b\)](#), [Bourguignon \(1979\)](#), and [Shorrocks \(1980\)](#). Yet this literature is often technical and dispersed, making it inaccessible to the applied researcher who most needs its insights.

Our contribution is to provide a unified, self-contained evaluation of inequality measures that is both rigorous and accessible. We define eleven properties that capture distinct intuitions about what inequality measurement should accomplish. We then evaluate eight indices against these properties, providing formal proofs where indices satisfy properties and concrete numerical counterexamples where they fail.¹ Throughout, we emphasize intuition:

¹We focus on the most widely used indices in applied economics. We exclude the Bonferroni index, Palma ratio, and optimal-transport-based indices, which are either closely related to indices we study (the Bonferroni to the Zenga, the Palma to percentile ratios) or too recent to have an established axiomatic literature.

each property is motivated by a real-world scenario that illustrates why it matters.

The theoretical part of the paper collects results that are mostly known and states them with self-contained proofs. The scale–translation impossibility (Theorem 23), the Shorrocks characterization of the GE family (Theorem 24), and the transfer–Lorenz equivalence (Theorem 26) are classical. The property-by-property evaluation draws on Atkinson (1970), Cowell (1977), Shorrocks (1980), and Shorrocks and Foster (1987). We add three things to this material. First, the Zenga index (Zenga, 2007) fails both diminishing transfer sensitivity and subgroup consistency; we give integer counterexamples (Propositions 18 and 20) and have not found either property settled in the literature on the index. Second, the three ways of applying Zenga’s definition to a finite population behave differently: an equal-weight average over split points cannot be population-invariant for any non-constant continuous pointwise function (Theorem 28); Zenga’s own finite-population formula is population-invariant but discontinuous where incomes tie and can rise after a progressive transfer; and the empirical-distribution version Z^* avoids both problems (Section 4.8). Third, we collect the known characterizations of decomposable indices (Shorrocks, 1980; Chakravarty and Tyagarupananda, 1998; Bosmans and Cowell, 2010) into a direct answer to whether counting properties favours relative measures: the maximum of our eleven properties is ten, reached by GE_α with $\alpha < 2$ and by the exponential absolute index alike (Theorem 30), so the choice between relative and absolute measurement has to be made on substantive grounds.

The empirical part asks a question the axiomatic literature usually leaves open: how often do these properties actually decide anything? We compute every index on 2,588 income and consumption distributions, each summarised by 100 percentile means, from the World Bank’s Poverty and Inequality Platform (PIP; World Bank, 2026a), and compare consecutive surveys within the platform’s comparable spells. The axioms make sharp predictions about these comparisons. When one Lorenz curve dominates the other, every relative (scale-invariant) index satisfying the transfer principle must move in the same direction (Theorem 26); when the curves cross once, the theorem of Shorrocks and Foster (1987) extends this agreement

to transfer-sensitive indices such as the MLD and the Theil index, under a condition on the coefficient of variation. Outside those cases the indices are free to disagree, and the question is how often they do.

Three findings stand out. First, Lorenz curves compared at every percentile cross in 62% of 1,955 survey pairs from 143 countries (36% if only the deciles are compared), so the unanimous rankings that the transfer principle delivers are often unavailable, as [Atkinson \(1970\)](#) and [Figini \(2000\)](#) found in smaller samples. When curves cross, seven relative indices that satisfy the transfer principle, one for each distinct ordering, agree on the direction of change in only 51% of pairs. Second, this disagreement is mostly about small changes: among crossing pairs in which the Gini moves by at least one point, the indices disagree in 23% of cases, and a noise screen based on each index’s own typical change gives a similar picture. Third, the transfer principle settles relative comparisons whenever Lorenz curves do not cross, but it does not settle the absolute question. Among the 458 pairs in which the Gini moves by at least one point and one Lorenz curve dominates, every relative index satisfying the transfer principle agrees, yet the absolute Gini moves in the opposite direction in 20% of them; over all such large changes the relative and absolute Gini conflict in 23% of pairs, against 9% for disagreement among relative indices. The reason is arithmetic: the absolute Gini μG rises whenever growth in the mean outpaces the proportional fall in the Gini, and in the PIP data it usually does, so the absolute Gini rose in 71% of the growth spells in which the Gini fell. This is the within-country counterpart of the debate over global inequality ([Ravallion, 2004](#); [Atkinson and Brandolini, 2010](#); [Niño Zarazúa et al., 2017](#)).

All computations are reproducible from the companion code.²

The remainder of the paper is organized as follows. Section 2 establishes the formal framework. Section 3 defines and motivates each property. Section 4 introduces the indices.

²Code repository: <https://github.com/diomavro/inequality-indices>. The library `inequality_indices.py` implements every index (the empirical code computes the 90/10 ratio from order statistics as in (17)); `verify_properties.py` and `verify_corrections.py` check the property table and the counterexamples; `pip_panel.py`, `pip_disagreement.py`, and `make_pip_figures.py` rebuild the empirical results, and every data-derived number in Section 7 is written to L^AT_EX by the code.

Section 5 evaluates each index against each property. Section 6 presents the characterization and impossibility results. Section 7 takes the indices to the PIP data. Section 8 offers practical guidance. Section 9 concludes.

2 Preliminaries

We consider a population of $n \geq 2$ individuals with positive incomes. An *income distribution* is a vector $\mathbf{y} = (y_1, y_2, \dots, y_n) \in \mathbb{R}_{++}^n$, where $\mathbb{R}_{++} = (0, \infty)$. We denote by

$$\mathcal{D} = \bigcup_{n=2}^{\infty} \mathbb{R}_{++}^n \quad (1)$$

the set of all possible income distributions of any finite population size.

For a distribution $\mathbf{y} \in \mathbb{R}_{++}^n$, we write $\mu(\mathbf{y}) = \frac{1}{n} \sum_{i=1}^n y_i$ for the arithmetic mean and $n(\mathbf{y}) = n$ for the population size. We use $y_{(1)} \leq y_{(2)} \leq \dots \leq y_{(n)}$ to denote the order statistics (sorted incomes).

Definition 1 (Inequality Index). An *inequality index* is a function $I : \mathcal{D} \rightarrow \mathbb{R}_{\geq 0}$ that assigns a non-negative real number to every income distribution.

Intuitively, $I(\mathbf{y})$ quantifies the degree of inequality in distribution $\mathbf{y}$. The higher the value, the more unequal the distribution. The restriction to non-negative values is natural: we want a measure that is minimized at perfect equality.

A key concept throughout is the *progressive transfer*, also known as a Pigou-Dalton transfer.

Definition 2 (Progressive Transfer). A *progressive transfer* from individual i to individual

j in distribution $\mathbf{y}$ is the operation that produces $\mathbf{y}'$ defined by

$$y'_k = \begin{cases} y_i - \delta & \text{if } k = i, \\ y_j + \delta & \text{if } k = j, \\ y_k & \text{otherwise,} \end{cases} \quad (2)$$

where $y_i > y_j$, $\delta > 0$, and $y_i - \delta \geq y_j + \delta$.

The last condition ensures that the transfer does not reverse the relative positions of the donor and recipient. Intuitively, we are taking from the rich and giving to the poor without making the recipient richer than the donor.

We also use the concept of a *k-fold replication*: for $\mathbf{y} \in \mathbb{R}_{++}^n$, the k -fold replication $\mathbf{y}^{(k)} \in \mathbb{R}_{++}^{kn}$ is the vector obtained by concatenating k copies of $\mathbf{y}$.

Finally, we recall the *Lorenz curve*. For a distribution $\mathbf{y} \in \mathbb{R}_{++}^n$ with order statistics $y_{(1)} \leq \dots \leq y_{(n)}$, the Lorenz curve $L_{\mathbf{y}} : [0, 1] \rightarrow [0, 1]$ is the piecewise linear function connecting the points

$$\left(\frac{k}{n}, \frac{\sum_{i=1}^k y_{(i)}}{\sum_{i=1}^n y_{(i)}} \right), \quad k = 0, 1, \dots, n. \quad (3)$$

The Lorenz curve plots the cumulative share of total income held by the bottom k/n fraction of the population. Perfect equality corresponds to the 45-degree line ($L(p) = p$ for all p), and greater inequality corresponds to a Lorenz curve that bows further below this line. We say that $\mathbf{y}$ *Lorenz-dominates* $\mathbf{z}$ if $L_{\mathbf{y}}(p) \geq L_{\mathbf{z}}(p)$ for all $p \in [0, 1]$ (with at least one strict inequality), meaning $\mathbf{y}$ is unambiguously more equal.

3 The Properties

We now define eleven properties that an inequality index might satisfy. For each, we provide the formal statement, an intuitive motivation, and—where helpful—a brief discussion of why the property might be controversial or why one might choose to relax it.

3.1 Anonymity

Property 1 (Anonymity). For any $\mathbf{y} \in \mathbb{R}_{++}^n$ and any permutation π of $\{1, \dots, n\}$,

$$I(y_{\pi(1)}, y_{\pi(2)}, \dots, y_{\pi(n)}) = I(\mathbf{y}). \quad (4)$$

Intuition. The identity of individuals should not matter—only the income values. If Alice earns \$30,000 and Bob earns \$80,000, inequality should be the same as if Bob earns \$30,000 and Alice earns \$80,000. This property reflects a commitment to treating individuals symmetrically and is universally accepted in the inequality measurement literature.

3.2 Normalization

Property 2 (Normalization). $I(\mathbf{y}) = 0$ if and only if $y_1 = y_2 = \dots = y_n$.

Intuition. Perfect equality—when everyone has the same income—should correspond to zero inequality, and this should be the *only* situation with zero inequality. This anchors the scale of the index and ensures that any deviation from perfect equality registers as positive inequality.

3.3 Continuity

Property 3 (Continuity). For each n , I is a continuous function on $\mathbb{R}_{++}^n$.

Intuition. Small changes in the income distribution should produce small changes in measured inequality. If one person’s income changes by one dollar, the inequality index should not jump discontinuously. This is a regularity condition that rules out pathological measures and ensures that measurement error in incomes does not lead to wild swings in inequality estimates.

3.4 Population Invariance

Property 4 (Population Invariance). For any $\mathbf{y} \in \mathbb{R}_{++}^n$ and any positive integer k ,

$$I(\mathbf{y}^{(k)}) = I(\mathbf{y}), \quad (5)$$

where $\mathbf{y}^{(k)}$ is the k -fold replication of $\mathbf{y}$.

Intuition. Cloning the entire population should not change inequality. If a country of 10 million people has a certain level of inequality, a country of 20 million with exactly the same income distribution—each person replicated—should have the same inequality. This property is essential for making meaningful comparisons across populations of different sizes.

3.5 Scale Invariance

Property 5 (Scale Invariance). For any $\mathbf{y} \in \mathbb{R}_{++}^n$ and any $\lambda > 0$,

$$I(\lambda y_1, \lambda y_2, \dots, \lambda y_n) = I(\mathbf{y}). \quad (6)$$

Intuition. Doubling everyone's income should not change inequality. If all incomes are expressed in euros rather than dollars, the inequality measure should not change. More substantively, this property encodes a *relative* view of inequality: what matters is not the absolute gaps between incomes but their ratios. A society where one person earns \$10,000 and another earns \$100,000 has the same inequality as one where they earn \$20,000 and \$200,000.

Remark 3. Scale invariance is sometimes called *mean independence* or *zero-degree homogeneity*. It implies that the inequality index depends only on relative incomes y_i/μ .

3.6 Translation Invariance

Property 6 (Translation Invariance). For any $\mathbf{y} \in \mathbb{R}_{++}^n$ and any $c \in \mathbb{R}$ such that $y_i + c > 0$ for all i ,

$$I(y_1 + c, y_2 + c, \dots, y_n + c) = I(\mathbf{y}). \quad (7)$$

Intuition. Adding \$1,000 to everyone’s income should not change inequality. This encodes an *absolute* view of inequality: what matters is the absolute gaps between incomes, not their ratios. Under this view, a society where one person earns \$10,000 and another earns \$20,000 (gap of \$10,000) is just as unequal as one where they earn \$110,000 and \$120,000 (same gap).

Most economists find the relative view (scale invariance) more natural for income inequality, but the absolute view has its defenders. [Kolm \(1976a\)](#) argued forcefully for “left-ist” (absolute) measures, and translation-invariant measures are sometimes preferred in the study of health inequality or wealth inequality.

The impossibility of combining scale and translation invariance has motivated a literature on *intermediate* inequality measures that satisfy neither fully but interpolate between the two views. [Bossert and Pfingsten \(1990\)](#) introduced the notion of a “unit-consistent” measure that satisfies a weakened form of both invariance properties, and [Zheng \(1994\)](#) studied the class of intermediate indices more broadly. These are available to researchers who find both the fully relative and fully absolute views unsatisfactory, but they fall outside the scope of our evaluation.

As we shall see in [Section 6](#), the relative and absolute views are fundamentally incompatible: no non-trivial continuous measure can satisfy both simultaneously.

3.7 The Pigou-Dalton Transfer Principle

Property 7 (Pigou-Dalton Transfer Principle). If $\mathbf{y}'$ is obtained from $\mathbf{y}$ by a progressive transfer ([Definition 2](#)), then $I(\mathbf{y}') < I(\mathbf{y})$.

Intuition. Taking from the rich and giving to the poor (without reversing their positions) should reduce inequality. This is arguably the most fundamental property of an inequality measure. If a transfer from Bill Gates to a minimum-wage worker does not reduce measured inequality, then the measure has failed at its most basic task.

The transfer principle was first articulated by [Pigou \(1912\)](#) and [Dalton \(1920\)](#).

Remark 4. The transfer principle is closely related to the concept of *Schur-convexity* in mathematics. A symmetric function is strictly Schur-convex if and only if it satisfies the transfer principle for mean-preserving transfers. Since progressive transfers preserve the mean, the transfer principle is equivalent to requiring that I be strictly Schur-convex when restricted to distributions with a given mean.

3.8 Lorenz Consistency

Property 8 (Lorenz Consistency). For any two distributions $\mathbf{y}, \mathbf{z} \in \mathbb{R}_{++}^n$ with $\mu(\mathbf{y}) = \mu(\mathbf{z})$, if $\mathbf{y}$ Lorenz-dominates $\mathbf{z}$, then $I(\mathbf{y}) < I(\mathbf{z})$.

Intuition. If the Lorenz curve of distribution $\mathbf{y}$ lies everywhere above that of $\mathbf{z}$ (closer to the line of perfect equality), then $\mathbf{y}$ should be judged less unequal than $\mathbf{z}$. Lorenz dominance represents the strongest possible unanimous ranking: every inequality-averse social welfare function would prefer $\mathbf{y}$ to $\mathbf{z}$.

The result of [Dasgupta et al. \(1973\)](#) establishes that for anonymous indices the transfer principle and Lorenz consistency are equivalent. The key insight is that Lorenz dominance can be achieved through a finite sequence of progressive transfers ([Hardy et al., 1934](#)), so any index satisfying the transfer principle must respect Lorenz dominance.

3.9 Diminishing Transfer Sensitivity

Property 9 (Diminishing Transfer Sensitivity). Consider two progressive transfers, each of size δ , in a distribution $\mathbf{y}$:

- (i) Transfer 1: from a person with income $y + d$ to a person with income y (at income level y).
- (ii) Transfer 2: from a person with income $y' + d$ to a person with income y' (at income level $y' > y$).

The reduction in inequality from Transfer 1 should be strictly greater than from Transfer 2.

Intuition. Transfers among the poor should matter more than transfers among the rich. A \$100 transfer from a person earning \$20,000 to a person earning \$10,000 should reduce inequality by more than a \$100 transfer from a person earning \$200,000 to a person earning \$190,000. Both transfers close the same absolute gap, but the first one occurs where the marginal utility of income is higher and where inequality is most keenly felt.

This property was formalized by [Shorrocks and Foster \(1987\)](#) and captures the idea that an inequality measure should give extra weight to what happens at the bottom of the distribution. Not all standard measures satisfy it, and its violation has practical consequences for how we evaluate anti-poverty versus middle-class-oriented redistributive policies.

3.10 Additive Decomposability

Property 10 (Additive Decomposability). For any partition of the population into $K \geq 2$ subgroups of at least two people each, with distributions $\mathbf{y}_1, \dots, \mathbf{y}_K$ (sizes $n_1, \dots, n_K$ and means $\mu_1, \dots, \mu_K$), the index can be written as

$$I(\mathbf{y}) = \sum_{k=1}^K w_k I(\mathbf{y}_k) + I(\mu_1 \mathbf{1}_{n_1}, \dots, \mu_K \mathbf{1}_{n_K}), \quad (8)$$

where $w_k = w_k(n_1, \dots, n_K, \mu_1, \dots, \mu_K)$ are weights depending only on subgroup sizes and means, and $\mathbf{1}_{n_k}$ is the n_k -vector of ones.

Intuition. Total inequality can be expressed as the sum of two components: (i) a weighted average of within-group inequalities, and (ii) between-group inequality, measured

as if every person received their group’s mean income. This decomposition allows researchers to answer questions like: “How much of total income inequality is due to differences *between* racial groups versus differences *within* racial groups?”

Additive decomposability is one of the most demanding properties on our list. As [Shorrocks \(1980\)](#) showed, it is highly restrictive: among scale-invariant measures satisfying standard axioms, only the Generalized Entropy family satisfies it.

3.11 Subgroup Consistency

Property 11 (Subgroup Consistency). Consider a partition of the population into K subgroups. If we replace the distribution in subgroup k with a new distribution $\mathbf{y}'_k$ of the same size n_k and same mean μ_k such that $I(\mathbf{y}'_k) > I(\mathbf{y}_k)$, while all other subgroups remain unchanged, then

$$I(\mathbf{y}'_1, \dots, \mathbf{y}'_k, \dots, \mathbf{y}_K) > I(\mathbf{y}_1, \dots, \mathbf{y}_k, \dots, \mathbf{y}_K). \quad (9)$$

Intuition. If inequality rises in one subgroup and nothing else changes, total inequality should rise. This is a minimal coherence requirement: we should not have a situation where inequality increases in, say, the South, while remaining unchanged in every other region, yet national inequality somehow *decreases*.

Subgroup consistency is weaker than additive decomposability: every additively decomposable index that satisfies the transfer principle is subgroup-consistent (Proposition 27), but not vice versa. [Shorrocks \(1988\)](#) showed that subgroup consistency, together with standard axioms and scale invariance, characterizes the class of indices that are ordinally equivalent to Generalized Entropy measures.

4 The Indices

We now define the eight inequality indices that we evaluate. For each, we provide the formula, a brief historical note, and a comment on its interpretation. Throughout, $\mathbf{y} = (y_1, \dots, y_n) \in$

$\mathbb{R}_{++}^n$ with mean $\mu = \mu(\mathbf{y})$.

4.1 Variance

Definition 5 (Variance).

$$\text{Var}(\mathbf{y}) = \frac{1}{n} \sum_{i=1}^n (y_i - \mu)^2. \quad (10)$$

The variance is the most elementary measure of dispersion in statistics. As an inequality index, it has the advantage of simplicity and additive decomposability. Its main drawback, as we shall see, is that it is not scale-invariant: doubling all incomes quadruples the variance. This makes it unsuitable for comparing inequality across countries or time periods with different income levels.

4.2 Coefficient of Variation

Definition 6 (Coefficient of Variation).

$$\text{CV}(\mathbf{y}) = \frac{\sqrt{\text{Var}(\mathbf{y})}}{\mu(\mathbf{y})} = \frac{1}{\mu} \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \mu)^2}. \quad (11)$$

The coefficient of variation normalizes the standard deviation by the mean, making it scale-invariant. It equals zero when all incomes are equal and increases without bound as inequality grows. The squared CV is equal to 2· the Generalized Entropy index with parameter $\alpha = 2$: $\text{CV}^2 = 2 \cdot GE_2$.

4.3 Variance of Logarithms

Definition 7 (Variance of Logarithms).

$$\text{VL}(\mathbf{y}) = \frac{1}{n} \sum_{i=1}^n \left(\ln y_i - \overline{\ln y} \right)^2, \quad (12)$$

where $\overline{\ln y} = \frac{1}{n} \sum_{i=1}^n \ln y_i$.

The variance of logarithms has been widely used in labor economics, partly because log wages are approximately normally distributed and partly because wage regressions are typically specified in logs (Juhn et al., 1993). Despite its popularity, this measure has a fundamental flaw: it can violate the Pigou-Dalton transfer principle, as we demonstrate in Section 5.

4.4 Gini Coefficient

Definition 8 (Gini Coefficient).

$$G(\mathbf{y}) = \frac{1}{2n^2\mu} \sum_{i=1}^n \sum_{j=1}^n |y_i - y_j|. \quad (13)$$

The Gini coefficient, introduced by Gini (1912), is the most widely used inequality measure. It equals twice the area between the Lorenz curve and the line of perfect equality, and it ranges between 0 (perfect equality) and $1 - 1/n$ (maximum inequality).³ An equivalent formulation uses the rank-weighted representation:

$$G(\mathbf{y}) = \frac{2}{n^2\mu} \sum_{i=1}^n \left(i - \frac{n+1}{2} \right) y_{(i)}, \quad (14)$$

where $y_{(1)} \leq y_{(2)} \leq \dots \leq y_{(n)}$ are the order statistics. This representation reveals that the Gini assigns linearly increasing weights to incomes based on their rank.

³Maximum inequality occurs when one person holds all the income: $\mathbf{y} = (0^+, \dots, 0^+, n\mu)$. In this case the sum of pairwise absolute differences is $2(n-1) \cdot n\mu$, giving $G = (n-1)/n = 1 - 1/n$. The Gini reaches 1 only in the limit as $n \rightarrow \infty$.

4.5 Generalized Entropy Family

Definition 9 (Generalized Entropy). For $\alpha \in \mathbb{R}$,

$$GE_\alpha(\mathbf{y}) = \begin{cases} \frac{1}{\alpha(\alpha-1)} \left[\frac{1}{n} \sum_{i=1}^n \left(\frac{y_i}{\mu} \right)^\alpha - 1 \right] & \text{if } \alpha \neq 0, 1, \\ \frac{1}{n} \sum_{i=1}^n \frac{y_i}{\mu} \ln \frac{y_i}{\mu} & \text{if } \alpha = 1 \quad (\text{Theil index}), \\ \frac{1}{n} \sum_{i=1}^n \ln \frac{\mu}{y_i} & \text{if } \alpha = 0 \quad (\text{Mean Log Deviation}). \end{cases} \quad (15)$$

The Generalized Entropy family, systematically studied by [Shorrocks \(1980\)](#) and [Cowell \(1977\)](#) (see also [Cowell, 2011](#)), is parameterized by α , which governs the measure's sensitivity to different parts of the income distribution. Lower values of α make the index more sensitive to changes at the bottom; higher values make it more sensitive to changes at the top. Two special cases are particularly important:

- **Theil index** ($\alpha = 1$): introduced by [Theil \(1967\)](#), it has an information-theoretic interpretation as the expected information content of the “income message.”
- **Mean Log Deviation** ($\alpha = 0$): also called the Theil- L index or Bourguignon index, it is the average percentage shortfall of individual income from the mean (in log terms).

The case $\alpha = 2$ gives $GE_2 = \frac{1}{2} CV^2$, connecting the GE family to the coefficient of variation.

4.6 Atkinson Index

Definition 10 (Atkinson Index). For inequality aversion parameter $\varepsilon > 0$,

$$A_\varepsilon(\mathbf{y}) = \begin{cases} 1 - \left[\frac{1}{n} \sum_{i=1}^n \left(\frac{y_i}{\mu} \right)^{1-\varepsilon} \right]^{1/(1-\varepsilon)} & \text{if } \varepsilon \neq 1, \\ 1 - \frac{\exp\left(\frac{1}{n} \sum_{i=1}^n \ln y_i\right)}{\mu} & \text{if } \varepsilon = 1. \end{cases} \quad (16)$$

The Atkinson index, introduced by [Atkinson \(1970\)](#), has a welfare-theoretic foundation. The quantity $y_e = (1 - A_\varepsilon)\mu$ is the *equally distributed equivalent income*: the per-capita income that, if given to everyone, would yield the same social welfare as the actual distribution. The parameter ε captures society's aversion to inequality. As $\varepsilon \rightarrow 0$, there is no inequality aversion and $A_\varepsilon \rightarrow 0$. As $\varepsilon \rightarrow \infty$, only the worst-off individual matters (Rawlsian case).

The Atkinson index is a monotone transformation of the GE family: A_ε is an increasing function of $GE_{1-\varepsilon}$. This means the two families give identical *rankings* of distributions, though they differ in cardinal terms.

4.7 Percentile Ratios

Definition 11 (Percentile Ratio). The p/q percentile ratio is

$$R_{p/q}(\mathbf{y}) = \frac{y_{(\lceil pn \rceil)}}{y_{(\lceil qn \rceil)}}, \quad (17)$$

where $y_{(k)}$ is the k -th order statistic and $0 < q < p < 1$.

The most common choices are the 90/10 ratio ($p = 0.9$, $q = 0.1$) and the 90/50 ratio. Percentile ratios are popular in the earnings inequality literature because they are robust to outliers and easy to interpret. However, they discard most of the information in the distribution, focusing only on two points. Equality corresponds to a ratio of 1, not 0, so

percentile ratios do not satisfy our normalization property (Property 2), which requires $I(\mathbf{y}) = 0$ at perfect equality.

4.8 Zenga Index

Zenga (2007) proposed an index built from the ratio of the mean income below a point of the distribution to the mean income above it. For a distribution with quantile function F^{-1} , write

$$M^-(p) = \frac{1}{p} \int_0^p F^{-1}(t) dt, \quad M^+(p) = \frac{1}{1-p} \int_p^1 F^{-1}(t) dt, \quad p \in (0, 1). \quad (18)$$

The *Zenga curve* $1 - M^-(p)/M^+(p)$ is the relative gap between the average income of the bottom $100p$ percent and that of the rest, and the Zenga index is its average over p . The index is less well known in anglophone economics than in the Italian and European statistics literature (Greselin et al., 2010, 2013; Langel and Tillé, 2012). Applying the definition to n incomes can be done in three ways, and they have different properties.

Definition 12 (Zenga Indices for a Finite Population). Let $y_{(1)} \leq \dots \leq y_{(n)}$ be sorted incomes, with $\bar{y}_i^-$ the mean of the i lowest and $\bar{y}_i^+$ the mean of the $n - i$ highest.

- (a) The *split-point average* gives each of the $n - 1$ split points equal weight:

$$Z(\mathbf{y}) = \frac{1}{n-1} \sum_{i=1}^{n-1} \left(1 - \frac{\bar{y}_i^-}{\bar{y}_i^+} \right). \quad (19)$$

- (b) *Zenga's finite-population index* (Zenga and Jędrzejczak, 2020, eq. 1) splits only at the distinct income values $x_1 < \dots < x_H$ and weights each split by the population share n_h/n at that value:

$$Z_N(\mathbf{y}) = \sum_{h=1}^{H-1} \frac{n_h}{n} \left(1 - \frac{\bar{y}(y_i \leq x_h)}{\bar{y}(y_i > x_h)} \right). \quad (20)$$

When all incomes are distinct, $Z_N = \frac{n-1}{n} Z$.

- (c) The *empirical Zenga index* applies the population definition to the empirical quantile function $F_{\mathbf{y}}^{-1}$:

$$Z^*(\mathbf{y}) = \int_0^1 \left(1 - \frac{M^-(p)}{M^+(p)} \right) dp. \quad (21)$$

All three are bounded in $[0, 1)$, equal zero if and only if all incomes are equal, and are scale-invariant. They differ on population invariance and continuity. The split-point average Z is continuous but not population-invariant: $Z(1, 3) \approx 0.667$ while $Z(1, 1, 3, 3) \approx 0.561$, and Theorem 28 shows that no equal-weight average of a non-constant continuous function of the ratios $\bar{y}_i^-/\bar{y}_i^+$ can be population-invariant. Zenga's own weighting solves this, $Z_N(1, 3) = Z_N(1, 1, 3, 3) = 1/3$, but at the cost of continuity, because tied incomes are merged into a single split: $Z_N(1, 1 + \epsilon, 3) \rightarrow 7/18$ as $\epsilon \rightarrow 0$, whereas $Z_N(1, 1, 3) = 4/9$. Z^* has both properties. Replicating the population leaves the empirical quantile function unchanged, $F_{\mathbf{y}^{(k)}}^{-1} = F_{\mathbf{y}}^{-1}$, so Z^* is population-invariant; and M^- and M^+ are integrals of the quantile function, which varies continuously with the incomes, so Z^* is continuous. Because $F_{\mathbf{y}}^{-1}$ is piecewise constant, Z^* can be computed as a sum of one-dimensional integrals over the intervals $((k-1)/n, k/n)$. The versions can differ substantially in small populations ($Z^*(1, 3) \approx 0.519$) but converge as n grows, and in the percentile data of Section 7 we use Z^* . The tie-merging in Z_N also breaks the transfer principle. A progressive transfer that makes two incomes equal merges two split points into one, and this can raise the index: moving 0.1 from the second to the first person in $(1, 1.2, 3)$ gives $(1.1, 1.1, 3)$ and raises Z_N from 0.386 to 0.422, while Z and Z^* fall. Z and Z^* satisfy the transfer principle; none of the three satisfies diminishing transfer sensitivity or subgroup consistency (Section 5).

5 Which Index Satisfies What?

We now systematically evaluate each index against each property. Results are summarized in Table 1.

Table 1: Axiomatic Properties of Inequality Indices

Property	Var	CV	VL	Gini	GE_α		A_ε	$R_{p/q}$	$Z / Z_N / Z^*$
					$\alpha < 2$	$\alpha \geq 2$			
Anonymity	✓	✓	✓	✓	✓	✓	✓	✓	✓
Normalization	✓	✓	✓	✓	✓	✓	✓	$\times^a$	✓
Continuity	✓	✓	✓	✓	✓	✓	✓	✓	$\checkmark / \times^b / \checkmark$
Population Inv.	✓	✓	✓	✓	✓	✓	✓	✓	$\times^b / \checkmark / \checkmark$
Scale Inv.	$\times$	✓	✓	✓	✓	✓	✓	✓	✓
Translation Inv.	✓	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$
Transfer Principle	✓	✓	$\times$	✓	✓	✓	✓	$\times$	$\checkmark / \times^b / \checkmark$
Lorenz Cons.	✓	✓	$\times$	✓	✓	✓	✓	$\times$	$\checkmark / \times / \checkmark$
Dim. Transfer Sens.	$\times$	$\times$	—	$\times$	✓	$\times$	✓	—	$\times^d$
Add. Decomposability	✓	$\times$	$\times$	$\times$	✓	✓	$\times$	$\times$	$\times$
Subgroup Cons.	✓	✓	$\times$	$\times$	✓	✓	✓	$\times$	$\times^d$

Notes: ✓ = satisfies; $\times$ = violates; — = not evaluated, because the index fails the transfer principle (Z_N also fails it, but a direct counterexample to DTS is given). ^aPercentile ratios equal 1 (not 0) at perfect equality. ^bZenga columns: split-point average Z / Zenga’s finite-population index Z_N / empirical index Z^* (Definition 12). Z fails population invariance (Theorem 28); Z_N is discontinuous where incomes tie and can rise after a transfer that creates a tie (Section 4.8). ^dAll three Zenga indices fail; integer counterexamples in Propositions 18 and 20. GE_α : Generalized Entropy with parameter α ; A_ε : Atkinson with parameter $\varepsilon > 0$. The GE column is split to show that GE_α with $\alpha < 2$ (including the Theil and MLD) satisfies every property except translation invariance (Theorem 30), while GE_α with $\alpha \geq 2$ fails diminishing transfer sensitivity.

Among all indices on our list, only GE_α with $\alpha < 2$ satisfies scale invariance, the transfer principle, DTS, additive decomposability, and subgroup consistency simultaneously; GE_0 (MLD) and GE_1 (Theil) are the most widely used members in this range.⁴

We now prove the entries that are not immediate from the definitions.

5.1 Anonymity

Proposition 13. *All eight indices satisfy anonymity.*

⁴ $GE_2 = \frac{1}{2}CV^2$, despite its popularity for top-income analysis, fails DTS and treats transfers among the poor identically to transfers among the rich. Researchers should choose α deliberately: lower α weights the bottom, higher α the top, but only $\alpha < 2$ ensures bottom-sensitivity.

Proof. Each index is defined as a symmetric function of the income vector: the variance, VL, and GE involve sums over individual incomes; the Gini involves a double sum of absolute differences; the Atkinson index involves a power-mean; percentile ratios depend on order statistics; and the Zenga index depends on order statistics through the lower and upper means. In each case, permuting the elements of $\mathbf{y}$ does not change the value of the index. $\square$

The universality of anonymity reflects its status as a minimal fairness requirement. An inequality measure that treated otherwise identical individuals differently based on their label would be difficult to justify.

5.2 Population Invariance

Proposition 14. *The Var, CV, VL, Gini, GE_α , A_ε , $R_{p/q}$, Z_N , and Z^* satisfy population invariance. The split-point Zenga average Z does not.*

Proof. The first seven indices depend only on summary statistics (means, ratios y_i/μ , ranks, or order statistics as percentile fractions) that are unchanged by exact population replication.

Zenga's Z_N (satisfies). Replication multiplies every frequency n_h and the population size by k , so the weights n_h/n and the lower and upper means at each distinct value are unchanged.

Empirical Zenga Z^* (satisfies). Population replication does not change the empirical quantile function: $F_{\mathbf{y}^{(k)}}^{-1} = F_{\mathbf{y}}^{-1}$. Hence $M^-(p)$, $M^+(p)$, and Z^* are all invariant.

Split-point average Z (fails). Z averages over $n - 1$ split points with equal weights. Replicating the population from n to kn changes the number of split points from $n - 1$ to $kn - 1$, altering the set of lower-upper mean ratios. *Counterexample:* $Z(1, 3) = 0.667$ but $Z(1, 1, 3, 3) = 0.561$. No non-trivial equal-weight average of a continuous function of the split-point ratios can satisfy population invariance (Theorem 28). $\square$

5.3 Scale Invariance

Proposition 15. *The CV, VL, Gini, GE_α (all α), A_ε (all $\varepsilon > 0$), $R_{p/q}$, and the three Zenga indices satisfy scale invariance. The variance does not.*

Proof. **Variance (fails).** $\text{Var}(\lambda \mathbf{y}) = \frac{1}{n} \sum_{i=1}^n (\lambda y_i - \lambda \mu)^2 = \lambda^2 \text{Var}(\mathbf{y})$. For $\lambda \neq 1$, this differs from $\text{Var}(\mathbf{y})$.

CV (satisfies). $\text{CV}(\lambda \mathbf{y}) = \frac{\lambda \sqrt{\text{Var}(\mathbf{y})}}{\lambda \mu} = \text{CV}(\mathbf{y})$.

VL (satisfies). $\ln(\lambda y_i) = \ln \lambda + \ln y_i$. The mean of logs shifts by $\ln \lambda$, so each deviation $\ln(\lambda y_i) - \overline{\ln(\lambda \mathbf{y})}$ equals $\ln y_i - \overline{\ln \mathbf{y}}$. Hence $\text{VL}(\lambda \mathbf{y}) = \text{VL}(\mathbf{y})$.

Gini (satisfies). $G(\lambda \mathbf{y}) = \frac{1}{2n^2 \lambda \mu} \sum_{i,j} |\lambda y_i - \lambda y_j| = \frac{\lambda}{2n^2 \lambda \mu} \sum_{i,j} |y_i - y_j| = G(\mathbf{y})$.

GE_α (satisfies). For $\alpha \neq 0, 1$: $GE_\alpha(\lambda \mathbf{y}) = \frac{1}{\alpha(\alpha-1)} \left[\frac{1}{n} \sum \left(\frac{\lambda y_i}{\lambda \mu} \right)^\alpha - 1 \right] = GE_\alpha(\mathbf{y})$. The cases $\alpha = 0$ and $\alpha = 1$ follow similarly, since the ratios y_i/μ are unchanged.

A_ε (satisfies). The Atkinson index depends only on the ratios y_i/μ , which are invariant to scaling.

$R_{p/q}$ (satisfies). $R_{p/q}(\lambda \mathbf{y}) = \frac{\lambda y_{(\lceil pn \rceil)}}{\lambda y_{(\lceil qn \rceil)}} = R_{p/q}(\mathbf{y})$.

Zenga (satisfies). Each ratio $\bar{y}_i^- / \bar{y}_i^+$ involves means of subsets of incomes. Scaling by λ multiplies both numerator and denominator by λ , so each ratio is unchanged, and hence $Z(\lambda \mathbf{y}) = Z(\mathbf{y})$. $\square$

The failure of the variance to satisfy scale invariance is its most serious practical limitation. If inequality in 1950 and 2020 is compared using the variance, real income growth alone will increase measured inequality even if the distribution has not changed in any relative sense.

5.4 Translation Invariance

Proposition 16. *The variance satisfies translation invariance. The CV, VL, Gini, GE_α , A_ε , $R_{p/q}$, and the Zenga indices do not.*

Proof. Variance (satisfies). $\text{Var}(\mathbf{y} + c\mathbf{1}) = \frac{1}{n} \sum_{i=1}^n (y_i + c - \mu - c)^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \mu)^2 = \text{Var}(\mathbf{y})$.

Others (fail). For $\mathbf{y} = (1, 3)$, $n = 2$, $\mu = 2$: $G = \frac{1}{2n^2\mu} \sum_{i,j} |y_i - y_j| = \frac{|1-1|+|1-3|+|3-1|+|3-3|}{2 \cdot 4 \cdot 2} = \frac{4}{16} = 0.25$.

Now add $c = 10$: $\mathbf{y}' = (11, 13)$, $\mu' = 12$. $G' = \frac{4}{2 \cdot 4 \cdot 12} = \frac{4}{96} \approx 0.042$. Since $G' \neq G$, the Gini is not translation-invariant. For the other normalized, continuous, scale-invariant indices, Theorem 23 shows that some uniform shift must change the index. The two indices on our list that fail normalization or continuity also fail translation invariance on this example: $R_{90/10}(1, 3) = 3$ but $R_{90/10}(11, 13) = 13/11$, and $Z_N(1, 3) = 1/3$ but $Z_N(11, 13) = 1/13$. $\square$

The variance is the only translation-invariant index in Table 1; other translation-invariant indices, such as the absolute Gini μG and the Kolm index, appear in Sections 6.6 and 7.

5.5 The Pigou-Dalton Transfer Principle

Proposition 17. *The Var, CV, Gini, GE_α (all α), A_ε (all $\varepsilon > 0$), Z , and Z^* satisfy the transfer principle. The VL, $R_{p/q}$, and Z_N do not.*

Proof. Variance (satisfies). Consider a progressive transfer of $\delta > 0$ from person i to person j , where $y_i > y_j$ and $y_i - \delta \geq y_j + \delta$. The mean is unchanged. The change in variance is

$$\begin{aligned} \Delta \text{Var} &= \frac{1}{n} \left[(y_i - \delta - \mu)^2 + (y_j + \delta - \mu)^2 - (y_i - \mu)^2 - (y_j - \mu)^2 \right] \\ &= \frac{1}{n} \left[-2\delta(y_i - \mu) + \delta^2 + 2\delta(y_j - \mu) + \delta^2 \right] \\ &= \frac{2\delta}{n} [\delta - (y_i - y_j)]. \end{aligned} \tag{22}$$

Since $y_i - y_j > 0$ and $\delta \leq (y_i - y_j)/2$, we have $\delta - (y_i - y_j) < 0$, so $\Delta \text{Var} < 0$.

CV (satisfies). Since the mean is unchanged and the variance decreases (by the argument above), $\text{CV} = \sqrt{\text{Var}}/\mu$ decreases.

Gini (satisfies). Using the rank-weighted representation (14), a progressive transfer that does not change ranks transfers income weight from a higher-ranked position to a lower-ranked position. Formally, if the donor has rank r_h and the recipient has rank $r_l < r_h$ (in ascending order), the change in the Gini is

$$\Delta G = -\frac{2\delta(r_h - r_l)}{n^2\mu} < 0. \quad (23)$$

This is negative since $r_h > r_l$, $\delta > 0$, and $\mu > 0$. A transfer that changes the ranks of third parties can be split into a finite sequence of smaller transfers between the same two people, each of which preserves ranks, so the Gini falls in that case too.

GE_α (**satisfies, all α**). We prove this using the convexity/concavity of the generating function.

Case $\alpha > 1$ or $\alpha < 0$: The function $g(y) = y^\alpha$ is strictly convex. For any strictly convex function g and a progressive transfer from y_i to y_j (with $y_i > y_j$, transfer amount δ , $y_i - \delta \geq y_j + \delta$):

$$g(y_i - \delta) + g(y_j + \delta) < g(y_i) + g(y_j), \quad (24)$$

which follows because the function's slope is steeper at y_i than at y_j , so the loss at the top exceeds the gain at the bottom. Since $GE_\alpha = \frac{1}{\alpha(\alpha-1)} \left[\frac{1}{n\mu^\alpha} \sum y_i^\alpha - 1 \right]$ and $\frac{1}{\alpha(\alpha-1)} > 0$ for $\alpha > 1$ or $\alpha < 0$, the index decreases.

Case $0 < \alpha < 1$: The function $g(y) = y^\alpha$ is strictly concave, so $\sum y_i^\alpha$ increases under a progressive transfer. But $\frac{1}{\alpha(\alpha-1)} < 0$, so the product decreases.

Case $\alpha = 1$ (Theil): The function $g(y) = y \ln y$ is strictly convex (since $g''(y) = 1/y > 0$), so $\sum y_i \ln y_i$ decreases. Since the Theil index equals $\frac{1}{n\mu} \sum y_i \ln y_i - \ln \mu$ and μ is unchanged, the Theil decreases.

Case $\alpha = 0$ (MLD): The function $g(y) = \ln y$ is strictly concave, so $\sum \ln y_i$ increases under a progressive transfer. Since $MLD = \ln \mu - \frac{1}{n} \sum \ln y_i$ and μ is unchanged, the MLD decreases.

A_ε (**satisfies**). Since A_ε is a monotonically increasing function of $GE_{1-\varepsilon}$, and $GE_{1-\varepsilon}$ satisfies the transfer principle, so does A_ε .

Zenga (satisfies). Consider a progressive transfer of δ from the individual at rank r_h to the individual at rank $r_l < r_h$ (in the sorted order), with δ small enough that ranks are preserved. We show each pointwise ratio $Z_i = 1 - \bar{y}_i^-/\bar{y}_i^+$ weakly decreases, with strict decrease for $r_l \leq i < r_h$.

Case $i < r_l$: Both individuals are in the upper group (ranks $i + 1, \dots, n$). The donor loses δ and the recipient gains δ , leaving the upper-group sum unchanged. The lower mean $\bar{y}_i^-$ involves only ranks $1, \dots, i$, none of which changed. Hence Z_i is unchanged.

Case $r_l \leq i < r_h$: The recipient (rank r_l) is in the lower group and the donor (rank r_h) is in the upper group. The lower mean increases: $\bar{y}_i^{-'} = \bar{y}_i^- + \delta/i$. The upper mean decreases: $\bar{y}_i^{+'} = \bar{y}_i^+ - \delta/(n - i)$. Both changes increase the ratio $\bar{y}_i^-/\bar{y}_i^+$, so $Z_i = 1 - \bar{y}_i^-/\bar{y}_i^+$ strictly decreases.

Case $i \geq r_h$: Both individuals are in the lower group. The sum of lower-group incomes is unchanged ($+\delta$ and $-\delta$ cancel), so $\bar{y}_i^-$ is unchanged. The upper mean involves neither individual, so $\bar{y}_i^+$ is unchanged. Hence Z_i is unchanged.

Since Z_i strictly decreases for every split point $i \in \{r_l, \dots, r_h - 1\}$ (a non-empty set) and is unchanged elsewhere, the Zenga index $Z = \frac{1}{n-1} \sum Z_i$ strictly decreases. For transfers large enough to change ranks, decompose into a finite sequence of rank-preserving transfers, each strictly reducing Z . For Z^* , a progressive transfer weakly raises the Lorenz curve L everywhere and strictly on an interval, so $M^-(p) = \mu L(p)/p$ weakly rises and $M^+(p) = \mu[1 - L(p)]/(1 - p)$ weakly falls at every p , strictly on a set of positive measure, and Z^* strictly falls. Z_N fails: the transfer from $(1, 1.2, 3)$ to $(1.1, 1.1, 3)$ raises it from 0.386 to 0.422 (Section 4.8).

VL (fails). *Counterexample*. Let $\mathbf{y} = (1, 2, 50, 100)$ and consider a progressive transfer of $\delta = 10$ from the person with income 100 to the person with income 50, yielding $\mathbf{y}' = (1, 2, 60, 90)$. Note that $100 - 10 = 90 \geq 60 = 50 + 10$, so this is a valid progressive transfer.

Computing the variance of logarithms:

$$\begin{aligned}\overline{\ln y} &= \frac{0 + 0.693 + 3.912 + 4.605}{4} = 2.303, \\ \text{VL}(\mathbf{y}) &= \frac{(0 - 2.303)^2 + (0.693 - 2.303)^2 + (3.912 - 2.303)^2 + (4.605 - 2.303)^2}{4} \\ &= \frac{5.302 + 2.590 + 2.590 + 5.302}{4} = 3.946.\end{aligned}$$

After the transfer:

$$\begin{aligned}\overline{\ln y'} &= \frac{0 + 0.693 + 4.094 + 4.500}{4} = 2.322, \\ \text{VL}(\mathbf{y}') &= \frac{(0 - 2.322)^2 + (0.693 - 2.322)^2 + (4.094 - 2.322)^2 + (4.500 - 2.322)^2}{4} \\ &= \frac{5.391 + 2.653 + 3.142 + 4.744}{4} = 3.982.\end{aligned}$$

Since $3.982 > 3.946$, the progressive transfer *increased* the variance of logarithms.

The logarithm is a concave function that “compresses” high incomes: the difference between $\ln(50)$ and $\ln(100)$ is only 0.693, while the difference between $\ln(1)$ and $\ln(2)$ is also 0.693, despite a much smaller absolute gap. When we transfer income between two high-income individuals, the transfer barely registers in log space. But the transfer changes the mean of logs, which shifts all the deviations, and this indirect effect can dominate, perversely increasing the variance of logarithms.

$R_{p/q}$ **(fails)**. Percentile ratios depend only on two order statistics. Any progressive transfer not involving the p -th or q -th percentile leaves the ratio unchanged. Moreover, a transfer between individuals both above the p -th percentile or both below the q -th percentile has no effect. $\square$

5.6 Diminishing Transfer Sensitivity

Proposition 18. GE_α satisfies diminishing transfer sensitivity if and only if $\alpha < 2$. The Atkinson index A_ε satisfies it for all $\varepsilon > 0$. The variance, CV, Gini, and the three Zenga

indices Z , Z_N , and Z^* do not satisfy it.

Proof. GE_α . For every α , $GE_\alpha(\mathbf{y}) = \frac{1}{n} \sum_i \phi_\alpha(y_i) + c$, where c and the strictly convex function ϕ_α depend on μ only, and $\phi_\alpha''(s) = s^{\alpha-2}/\mu^\alpha$ (for $\alpha = 0$, $\phi_0(s) = -\ln s$; for $\alpha = 1$, $\phi_1(s) = (s/\mu) \ln s$ up to terms linear in s). A transfer of δ from $y+d$ to y leaves μ unchanged and reduces $n GE_\alpha$ by

$$\Delta_\alpha(y) = \phi_\alpha(y) + \phi_\alpha(y+d) - \phi_\alpha(y+\delta) - \phi_\alpha(y+d-\delta) = \int_0^\delta \int_t^{d-\delta+t} \phi_\alpha''(y+s) ds dt, \quad (25)$$

which holds for any transfer size. The region of integration does not depend on y , so $\Delta_\alpha(y)$ is strictly decreasing in y when ϕ_α'' is strictly decreasing ($\alpha < 2$), constant when $\alpha = 2$, and strictly increasing when $\alpha > 2$. Hence GE_α satisfies DTS if and only if $\alpha < 2$, for every transfer size $\delta \leq d/2$.

Variance and CV (fail). From equation (22), the change in variance from a transfer of δ with gap $d = y_i - y_j$ is $\frac{2\delta}{n}[\delta - d]$, which depends on d and δ but *not* on the income level y_j . Since $CV^2 = 2 \cdot GE_2$ and GE_2 has $\alpha = 2$ (the level-neutral case), the CV also fails DTS.

Gini (fails). From equation (23), $|\Delta G| = \frac{2\delta(r_h - r_l)}{n^2\mu}$, which depends on the *rank gap* $r_h - r_l$, not on income levels. Counterexample: consider $\mathbf{y} = (1, 2, 3, 98, 99, 100)$. A transfer of $\delta = 0.5$ from income 2 to income 1 has rank gap 1. A transfer of $\delta = 0.5$ from income 99 to income 98 also has rank gap 1. Both reduce the Gini by the same amount, violating DTS.

A_ε (satisfies). Since A_ε is a monotone increasing function of $GE_{1-\varepsilon}$ and $1-\varepsilon < 1 < 2$ for all $\varepsilon > 0$, the Atkinson index inherits DTS from $GE_{1-\varepsilon}$. To see why a monotone increasing transformation $A_\varepsilon = \phi(GE_{1-\varepsilon})$ preserves DTS: a progressive transfer that reduces $GE_{1-\varepsilon}$ by a larger amount at income level y than at $y' > y$ also reduces $A_\varepsilon = \phi(GE_{1-\varepsilon})$ by a larger amount at y (since ϕ is increasing, larger reductions in the argument produce larger reductions in ϕ). Hence the relative ranking of transfer effects on $GE_{1-\varepsilon}$ is preserved under ϕ , and DTS carries over to A_ε .

Zenga (fails). Take $\mathbf{y} = (3, 6, 7, 7, 8)$ and two transfers of $\delta = 1/2$ across a gap of $d = 1$: the low transfer from 7 to 6 and the high transfer from 8 to 7. DTS requires the low transfer to reduce inequality by more. It reduces Z , Z_N , and Z^* by 0.0123, 0.0180, and 0.0102, whereas the high transfer reduces them by 0.0161, 0.0547, and 0.0171. For comparison, the MLD falls by 0.0012 after the low transfer and by 0.0009 after the high one, as DTS requires. A transfer between adjacent ranks r and $r + 1$ changes only the split-point term at r ; to first order the reduction in Z is proportional to $[1/r + (1 - Z_r)/(n - r)]/\bar{y}_r^+$, with $Z_r = 1 - \bar{y}_r^-/\bar{y}_r^+$. The second term grows as r approaches n , so transfers near the top of the distribution can outweigh transfers near the bottom. In simulations with log-normal incomes the high transfer still has the larger effect in about 2% of random comparisons at $n = 100$ (companion code), so the failure does not disappear in large populations. $\square$

The economic implications of this result are significant. An index without diminishing transfer sensitivity treats a \$100 transfer between millionaires as equally inequality-reducing as a \$100 transfer between minimum-wage workers. For researchers studying poverty-related inequality or evaluating progressive tax policies, choosing an index with DTS (such as Theil, MLD, or Atkinson) is essential.

5.7 Additive Decomposability

Proposition 19. *The variance and GE_α (all α) satisfy additive decomposability. The CV, VL, Gini, A_ε , $R_{p/q}$, and the Zenga indices do not.*

Proof. **Variance (satisfies).** The standard ANOVA decomposition gives

$$\text{Var}(\mathbf{y}) = \sum_{k=1}^K \frac{n_k}{n} \text{Var}(\mathbf{y}_k) + \sum_{k=1}^K \frac{n_k}{n} (\mu_k - \mu)^2, \quad (26)$$

where the first term is the population-weighted average of within-group variances and the second term is the variance of group means—which equals $\text{Var}(\mu_1 \mathbf{1}_{n_1}, \dots, \mu_K \mathbf{1}_{n_K})$. This fits the form of equation (8) with weights $w_k = n_k/n$.

GE_α (satisfies). For all α ,

$$GE_\alpha(\mathbf{y}) = \sum_{k=1}^K \frac{n_k}{n} \left(\frac{\mu_k}{\mu} \right)^\alpha GE_\alpha(\mathbf{y}_k) + GE_\alpha(\mu_1 \mathbf{1}_{n_1}, \dots, \mu_K \mathbf{1}_{n_K}). \quad (27)$$

The weights are $w_k = \frac{n_k}{n} \left(\frac{\mu_k}{\mu} \right)^\alpha$. For $\alpha = 0$ (MLD), $w_k = n_k/n$ (population shares). For $\alpha = 1$ (Theil), $w_k = \frac{n_k}{n} \frac{\mu_k}{\mu}$ (income shares). For $\alpha = 2$, $w_k = \frac{n_k}{n} \left(\frac{\mu_k}{\mu} \right)^2$.

To verify, consider the case $\alpha = 0$ (MLD):

$$\begin{aligned} MLD(\mathbf{y}) &= \frac{1}{n} \sum_{i=1}^n \ln \frac{\mu}{y_i} = \frac{1}{n} \sum_{k=1}^K \sum_{i \in k} \left(\ln \frac{\mu}{\mu_k} + \ln \frac{\mu_k}{y_i} \right) \\ &= \sum_{k=1}^K \frac{n_k}{n} \ln \frac{\mu}{\mu_k} + \sum_{k=1}^K \frac{n_k}{n} \cdot \frac{1}{n_k} \sum_{i \in k} \ln \frac{\mu_k}{y_i} \\ &= MLD(\mu_1 \mathbf{1}_{n_1}, \dots, \mu_K \mathbf{1}_{n_K}) + \sum_{k=1}^K \frac{n_k}{n} MLD(\mathbf{y}_k). \end{aligned} \quad (28)$$

A test for failure. In (8) the weights and the between-group term depend only on subgroup sizes and means. So if two versions of one subgroup have the same size, mean, and index value, and the other subgroups are unchanged, an additively decomposable index must give the same total. Exhibiting two such versions with different totals shows that no choice of weights works.

Gini (fails). With $A = (3, 9)$, the subgroups $B = (2, 4, 6)$ and $B' = (2.6, 2.8, 6.6)$ both have mean 4 and Gini $2/9$, but $G(A, B) = 0.283$ and $G(A, B') = 0.277$. The Gini does decompose exactly when subgroups do not overlap in income; the failure comes from the overlap term, which depends on how the subgroup distributions interleave ([Bhattacharya and Mahalanobis, 1967](#)).

$R_{p/q}$ (fails). For the 90/10 ratio, $A = (4, 14)$ with $B = (3, 5, 18)$ or $B' = (2, 12, 12)$: both versions of B have mean $26/3$ and ratio 6, but the ratio for the whole population is 6 in the first case and 7 in the second.

Zenga (fails). The same construction works for Z , Z_N , and Z^* : with $A = (3, 9)$ and $B = (2, 4, 6)$, a second version of B with the same mean and the same index value, of the

form $(x, x + 0.2, 11.8 - 2x)$, changes the total index in each case (companion code).

VL and A_ε (fail). Each is scale-invariant and satisfies the other conditions of Theorem 24, including differentiability, but neither is a positive multiple of a GE index (A_ε is a nonlinear transformation of $GE_{1-\varepsilon}$, and the VL is not a function of any single GE index), so by that theorem neither is additively decomposable.

CV (fails). The CV is not differentiable at perfect equality, so we argue directly. Take two subgroups with fixed sizes and means and vary the CV c_1 of the first. Since $CV^2 = 2GE_2$ is additively decomposable, the total CV equals $\sqrt{ac_1^2 + b}$ for constants $a > 0$ and $b > 0$ that do not depend on c_1 . Additive decomposability of the CV would require the total to be affine in c_1 , but $\sqrt{ac_1^2 + b}$ is strictly convex in c_1 whenever $b > 0$, for instance whenever the subgroup means differ. $\square$

A researcher studying the evolution of income inequality over time wants to know whether changes are driven by widening gaps between demographic groups (between-group) or by increasing dispersion within groups (within-group). Only additively decomposable measures provide a clean answer.

5.8 Subgroup Consistency

Proposition 20. *The Var, CV, GE_α (all α), and A_ε (all $\varepsilon > 0$) satisfy subgroup consistency. The Gini, VL, $R_{p/q}$, and the three Zenga indices do not.*

Proof. **Var and GE_α (satisfy).** These indices are additively decomposable. In the decomposition (8), if the distribution in subgroup k changes to $\mathbf{y}'_k$ with the same mean μ_k and $I(\mathbf{y}'_k) > I(\mathbf{y}_k)$, the between-group component is unchanged (since all group means and sizes are unchanged), and the within-group component increases (since $w_k > 0$ and the k -th term increases). Hence total inequality increases.

CV (satisfies). $CV^2 = 2 \cdot GE_2$ is additively decomposable and satisfies subgroup consistency. Since the CV is a monotone increasing function of CV^2 , it also satisfies subgroup

consistency.

A_ε (satisfies). A_ε is a monotone increasing function of $GE_{1-\varepsilon}$. If within-group inequality increases (as measured by A_ε in subgroup k), then $GE_{1-\varepsilon}$ in subgroup k also increases (same monotone relationship). By subgroup consistency of $GE_{1-\varepsilon}$, total $GE_{1-\varepsilon}$ increases, and hence total A_ε increases.

Zenga (fails). Let subgroup $A = (36, 18, 15)$ and subgroup $B = (28, 23, 2, 3)$, and replace A by $A' = (33, 12, 24)$, which has the same size and mean (23). Within the subgroup, Z^* rises from 0.4356 to 0.4589. For the whole population, Z^* falls from 0.7253 to 0.7122. The MLD rises in both the subgroup and the total, as subgroup consistency requires. The same example works for the other two versions: Z rises from 0.4931 to 0.5167 in the subgroup and falls from 0.7372 to 0.7252 in total, and Z_N rises from 0.3287 to 0.3445 and falls from 0.6319 to 0.6216. Random searches over small integer populations find violations in about 2% of same-mean replacements for Z and Z^* (companion code).

Gini (fails). The failure is consistent with the characterization of [Shorrocks \(1988\)](#), under which subgroup consistency together with the other standard axioms and scale invariance characterizes indices ordinally equivalent to a GE index, but a direct example is simpler. A small example: replace subgroup $A = (19, 8, 8)$ by $A' = (13, 5, 17)$, which has the same size and mean, leaving $B = (15, 14, 13)$ unchanged. The Gini of the subgroup rises from 0.210 to 0.229, but the Gini of the whole population falls from 0.167 to 0.145.

VL (fails). Replace subgroup $A = (16, 16)$ by $A' = (18, 14)$, leaving $B = (2, 2)$ unchanged. The VL of the subgroup rises from 0 to 0.0158, but the VL of the whole population falls from 1.0810 to 1.0807.

$R_{p/q}$ (fails). Changes in the interior of a subgroup's distribution may not affect the overall percentile ratio, so increasing inequality within a subgroup can leave the overall $R_{p/q}$ unchanged. $\square$

The failure of the Gini to satisfy subgroup consistency is one of its most significant limitations. It means that the Gini can produce paradoxical results: inequality could increase

in every region of a country, yet the national Gini could decrease due to changes in how regional distributions overlap.

5.9 Lorenz Consistency

Proposition 21. *The Var, CV, Gini, GE_α , A_ε , Z , and Z^* satisfy Lorenz consistency. The VL, $R_{p/q}$, and Z_N do not.*

Proof. By Theorem 26, an anonymous index is Lorenz-consistent if and only if it satisfies the transfer principle. The indices in the first list satisfy the transfer principle (Proposition 17 and Section 4.8); the VL, $R_{p/q}$, and Z_N do not. $\square$

Remark 22. For indices that are also scale-invariant (CV, Gini, GE_α , A_ε), Lorenz consistency extends to distributions with different means: if $L_{\mathbf{y}}(p) \geq L_{\mathbf{z}}(p)$ for all p , then $I(\mathbf{y}) \leq I(\mathbf{z})$, even when $\mu(\mathbf{y}) \neq \mu(\mathbf{z})$. The variance, lacking scale invariance, is Lorenz-consistent only for same-mean comparisons.

6 Characterization and Impossibility Results

We now present two fundamental results that deepen our understanding of the landscape of inequality measures.

6.1 The Impossibility of Reconciling Relative and Absolute Views

Theorem 23 (Scale–Translation Impossibility). *There exists no inequality index $I : \mathcal{D} \rightarrow \mathbb{R}_{\geq 0}$ that simultaneously satisfies normalization (Property 2), continuity (Property 3), scale invariance (Property 5), and translation invariance (Property 6).*

Intuition. The relative view says that inequality is about ratios: \$10,000 vs. \$100,000 is the same as \$1 vs. \$10. The absolute view says inequality is about gaps: a \$90,000 gap is a

\$90,000 gap regardless of the income level. The two views disagree whenever incomes grow or shrink by more than a uniform amount, and no continuous measure can agree with both.

Proof. Suppose for contradiction that such an index I exists. Consider two-person distributions $\mathbf{y} = (1, x)$ for $x > 1$, and define $f(x) = I(1, x)$. By normalization, $f(1) = 0$ and $f(x) > 0$ for $x > 1$.

We show that f must be constant on $(1, \infty)$, contradicting normalization.

Fix any $x > 1$ and $x' > 1$. Choose $c = \frac{x-x'}{x'-1}$. We verify that $c > -1$ (so the translation is valid) and that both components remain positive:

- $1 + c = \frac{x-1}{x'-1} > 0$ since $x > 1$ and $x' > 1$.
- $x + c = \frac{x(x'-1)+x-x'}{x'-1} = \frac{xx'-x'}{x'-1} = \frac{x'(x-1)}{x'-1} > 0$.

Now apply translation invariance followed by scale invariance:

$$\begin{aligned} f(x) &= I(1, x) = I(1 + c, x + c) && \text{(translation invariance)} \\ &= I\left(1, \frac{x + c}{1 + c}\right) && \text{(scale invariance, dividing by } 1 + c > 0\text{)}. \end{aligned} \quad (29)$$

Computing the second argument:

$$\frac{x + c}{1 + c} = \frac{x + \frac{x-x'}{x'-1}}{\frac{x-1}{x'-1}} = \frac{\frac{x(x'-1)+x-x'}{x'-1}}{\frac{x-1}{x'-1}} = \frac{x'(x-1)}{x-1} = x'. \quad (30)$$

Therefore $f(x) = f(x')$ for any $x, x' > 1$. So f is constant on $(1, \infty)$; call this constant c_0 .

By normalization, $f(1) = 0$ and $c_0 > 0$ (since $f(x) > 0$ for $x > 1$). But then $\lim_{x \rightarrow 1^+} f(x) = c_0 > 0 \neq f(1) = 0$, contradicting continuity. $\square$

6.2 The Generalized Entropy Characterization

Theorem 24 (Shorrocks Characterization, [Shorrocks, 1980](#)). *An inequality index $I : \mathcal{D} \rightarrow \mathbb{R}_{\geq 0}$ satisfies anonymity (Property 1), normalization (Property 2), continuity (Property 3), population invariance (Property 4), scale invariance (Property 5), and additive decomposability (Property 10), and has continuous first-order partial derivatives, if and only if I is a positive scalar multiple of GE_α for some $\alpha \in \mathbb{R}$.*

Intuition. The power of this theorem lies in what it rules out. If a researcher wants an inequality measure that is scale-invariant, can compare populations of different sizes, behaves continuously, and can be decomposed into within-group and between-group components, then the *only* options are the GE indices. The Atkinson index and the variance of logarithms, which satisfy every other condition, are therefore not additively decomposable. The Gini is excluded as well (Proposition 19), although it falls outside the theorem because it is not differentiable where incomes tie.

We state this result without proof; the full argument is in [Shorrocks \(1980\)](#). The key step is that additive decomposability and scale invariance force the index to satisfy the functional equation $I(\mathbf{y}) = \sum_k w_k(\mu_k, n_k, \mu, n) I(\mathbf{y}_k) + I(\mu_1 \mathbf{1}_1, \dots, \mu_K \mathbf{1}_K)$ with specific weight structure. The differentiability assumption rules out irregular solutions of the resulting functional equations.

Remark 25 (Dropping One Axiom). Four of the axioms in Theorem 24 cannot be dropped without admitting indices outside the GE family.

- (i) *Normalization.* The index $I \equiv 0$ satisfies every other axiom but is zero for unequal distributions.
- (ii) *Population Invariance.* $I_n(\mathbf{y}) = \frac{n-1}{n} GE_1(\mathbf{y})$ satisfies the other axioms, with decomposition weights rescaled by $\frac{n-1}{n} \big/ \frac{n_k-1}{n_k}$. But $I_2(1, 3) = \frac{1}{2} \cdot 0.1308 = 0.0654$ while $I_4(1, 1, 3, 3) = \frac{3}{4} \cdot 0.1308 = 0.0981$.

(iii) *Scale Invariance.* The variance satisfies the other axioms but $\text{Var}(2, 6) = 4 \neq 1 = \text{Var}(1, 3)$.

(iv) *Additive Decomposability.* The Atkinson index A_ϵ satisfies the other axioms, including differentiability, and is not a multiple of any GE_α , so by the theorem it is not additively decomposable.

6.3 The Transfer–Lorenz Equivalence

Theorem 26 (Dasgupta–Sen–Starrett, [Dasgupta et al., 1973](#)). *For an anonymous inequality index I , the following are equivalent:*

- (i) *I satisfies the Pigou–Dalton transfer principle (Property 7).*
- (ii) *I is Lorenz-consistent (Property 8).*

Intuition. This theorem tells us that the transfer principle and Lorenz consistency are really two ways of saying the same thing. The transfer principle is a “local” condition (one pairwise transfer), while Lorenz consistency is a “global” condition (the entire curve lies above). The bridge between them is the Hardy–Littlewood–Pólya theorem, which shows that any Lorenz improvement can be decomposed into elementary progressive transfers.

Proof. (i) $\Rightarrow$ (ii). Suppose $\mathbf{y}$ Lorenz-dominates $\mathbf{z}$ (with the same mean and population size). By the Hardy–Littlewood–Pólya theorem ([Hardy et al., 1934](#)), there exists a finite sequence of progressive transfers transforming $\mathbf{z}$ into $\mathbf{y}$. Since $\mathbf{y}$ and $\mathbf{z}$ are not permutations of each other, the sequence contains at least one transfer; each transfer strictly reduces I (by the transfer principle) and permutations leave it unchanged (by anonymity), so $I(\mathbf{y}) < I(\mathbf{z})$.

(ii) $\Rightarrow$ (i). If $\mathbf{y}$ is obtained from $\mathbf{z}$ by a progressive transfer, then $L_{\mathbf{y}}(p) \geq L_{\mathbf{z}}(p)$ for all p , with strict inequality at some p , because the bottom group that contains the recipient but not the donor gains δ . So $\mathbf{y}$ Lorenz-dominates $\mathbf{z}$, and Lorenz consistency gives $I(\mathbf{y}) < I(\mathbf{z})$. $\square$

6.4 Which Properties Can Coexist?

The results of this section reveal not only individual properties of specific indices, but a deeper structure: some pairs of desirable properties are logically incompatible, while others are not only compatible but mutually reinforcing. We organise these relationships in Proposition 27 and Table 2.

Proposition 27 (Property Compatibility). *Among anonymous, normalised, continuous, population-invariant inequality indices:*

- (i) **Scale invariance and translation invariance are incompatible.** *No such index satisfies both simultaneously (Theorem 23).*
- (ii) **Additive decomposability and scale invariance jointly force the GE family.** *Any index satisfying scale invariance and additive decomposability (in addition to the four standard axioms) must be a positive multiple of GE_α for some α (Theorem 24). For example, the Gini is scale-invariant but not decomposable, and the variance is decomposable but not scale-invariant.*
- (iii) **Additive decomposability implies subgroup consistency.** *Any additively decomposable index satisfying the transfer principle is subgroup-consistent.*

Proof. Let I be additively decomposable with decomposition (8), and suppose within-group k inequality increases: $I(\mathbf{y}'_k) > I(\mathbf{y}_k)$, with $\mu(\mathbf{y}'_k) = \mu(\mathbf{y}_k)$ and all other groups unchanged. Since the weights w_k depend only on subgroup sizes and means, all weights are unchanged. The between-group term is also unchanged (group means are unchanged). Therefore

$$I(\mathbf{y}') = \sum_{j \neq k} w_j I(\mathbf{y}_j) + w_k I(\mathbf{y}'_k) + I_{\text{between}} > \sum_j w_j I(\mathbf{y}_j) + I_{\text{between}} = I(\mathbf{y}),$$

provided $w_k > 0$. The weights are indeed positive: a progressive transfer inside group k changes neither the group means nor the other groups, so by (8) it changes I by w_k

times the change in $I(\mathbf{y}_k)$; the transfer principle requires both changes to be negative, so $w_k > 0$. $\square$

The converse does not hold: A_ε is subgroup-consistent but not additively decomposable.

(iv) **The transfer principle and Lorenz consistency are equivalent.** (Theorem 26.)

Any index satisfying one satisfies the other; they cannot be separated.

(v) **Translation invariance is compatible with the transfer principle, diminishing transfer sensitivity, and additive decomposability.** *The exponential absolute index E_κ of Theorem 30 satisfies all three. What translation invariance rules out is scale invariance, by (i).*

Table 2: Pairwise Compatibility of Key Properties

	Scale Inv.	Trans. Inv.	Transfer Pr.	Add. Decomp.	Subgr. Cons.
Scale Inv.	—	NO	Yes	Yes*	Yes
Trans. Inv.	NO	—	Yes [†]	Yes [†]	Yes [†]
Transfer Pr.	Yes	Yes [†]	—	Yes	Yes
Add. Decomp.	Yes*	Yes [†]	Yes	—	Yes
Subgr. Cons.	Yes	Yes [†]	Yes	Yes	—

Notes: “Yes” = jointly satisfiable by at least one index; “**NO**” = incompatible for any continuous normalised index (Theorem 23). *Scale invariance + additive decomposability forces the GE family (Theorem 24). [†]Attained by the variance and by the exponential absolute index E_κ (Theorem 30); translation invariance is incompatible only with scale invariance.

The compatibility structure has a direct practical implication. A researcher who requires decomposability for structural analysis effectively has no choice among scale-invariant measures: the GE family is uniquely determined by Theorem 24. A researcher who needs only subgroup consistency has a slightly wider menu: the Atkinson index also qualifies, but the

Gini and the Zenga indices do not (Proposition 20). A researcher who adopts the absolute view (translation invariance) gives up scale invariance but nothing else: the exponential index E_κ is decomposable and bottom-sensitive, and the Kolm index is its ordinal equivalent.

6.5 Impossibility for Split-Point Indices

The split-point average Z fails population invariance (Section 4.8). One might hope that a different choice of pointwise function could fix this. The following theorem shows that it cannot: *any* index that averages a continuous function of the conditional-mean ratios over the $n - 1$ split points with equal weights must be trivial if it is to satisfy population invariance.

Theorem 28 (Split-Point Impossibility). *Let $f : (0, \infty) \rightarrow \mathbb{R}$ be continuous with $f(1) = 0$, and define*

$$I_f(\mathbf{y}) = \frac{1}{n-1} \sum_{i=1}^{n-1} f\left(\frac{\bar{y}_i^-}{\bar{y}_i^+}\right). \quad (31)$$

If I_f satisfies population invariance, then $f \equiv 0$ on $(0, 1]$, so I_f is identically zero.

Intuition. Population replication creates new split points that “interpolate” between the original ones. The averaging denominator changes from $n - 1$ to $kn - 1$, and the new pointwise values at the interpolated split points cannot exactly compensate. The proof shows that the functional equation generated by 2-fold replication forces f to be its own midpoint average on orbits that converge to the fixed point $r = 1$; continuity and $f(1) = 0$ then force $f \equiv 0$.

Proof. Consider $\mathbf{y} = (a, b)$ with $0 < a < b$ and $r = a/b \in (0, 1)$. Then $I_f(a, b) = f(r)$. The 2-fold replication $\mathbf{y}^{(2)} = (a, a, b, b)$ has three split points with ratios

$$\phi(r) = \frac{3r}{r+2}, \quad r, \quad \psi(r) = \frac{2r+1}{3}.$$

Population invariance gives $f(r) = \frac{1}{3}[f(\phi(r)) + f(r) + f(\psi(r))]$, i.e.,

$$2f(r) = f(\phi(r)) + f(\psi(r)). \quad (32)$$

For $r \in (0, 1)$, both $\phi(r)$ and $\psi(r)$ lie strictly between r and 1: they are contractions toward the fixed point $r = 1$.

Fix any $\delta \in (0, 1)$ and consider f on $[\delta, 1]$. Since f is continuous on a compact set, $M := \max_{r \in [\delta, 1]} |f(r)|$ is attained at some $r^* \in [\delta, 1)$ (the case $r^* = 1$ gives $M = 0$, and we are done). From (32), $f(r^*) = \frac{1}{2}[f(\phi(r^*)) + f(\psi(r^*))]$. Since $\phi(r^*), \psi(r^*) \in (r^*, 1) \subseteq [\delta, 1]$ and $|f| \leq M$ on this interval, the average can equal $\pm M$ only if $f(\phi(r^*)) = f(\psi(r^*)) = f(r^*)$. Iterating with ψ : $f(\psi^k(r^*)) = f(r^*)$ for all $k \geq 1$. But $\psi^k(r^*) \rightarrow 1$ and $f(1) = 0$, so by continuity $f(r^*) = 0$, contradicting $|f(r^*)| = M > 0$. Therefore $M = 0$ and $f \equiv 0$ on $[\delta, 1]$.

Since $\delta > 0$ was arbitrary, $f \equiv 0$ on $(0, 1]$. Every ratio $\bar{y}_i^-/\bar{y}_i^+$ of sorted incomes lies in $(0, 1]$, so I_f is identically zero; the values of f above 1 never enter the index. $\square$

Remark 29. The theorem locates the problem in the equal weights, and the two other versions of Definition 12 escape it in different ways. Zenga's finite-population index Z_N weights each distinct income value by its population share, which restores population invariance but makes the index discontinuous where incomes tie. The empirical index Z^* integrates over all $p \in (0, 1)$; since the empirical quantile function is invariant under replication, it is population-invariant and continuous.

6.6 Axiomatic Optimality Bounds

How many of the eleven properties can one index satisfy? The answer is ten, and it is the same for relative and absolute indices. For the absolute side we need the exponential index

$$E_\kappa(\mathbf{y}) = \frac{1}{n} \sum_{i=1}^n e^{\kappa(\mu - y_i)} - 1, \quad \kappa > 0, \quad (33)$$

which is an increasing transformation of the Kolm index $K_\kappa(\mathbf{y}) = \kappa^{-1} \ln[1 + E_\kappa(\mathbf{y})]$ (Kolm, 1976a). E_κ stands to K_κ as $GE_{1-\varepsilon}$ stands to A_ε : the two rank distributions identically, but only the first is additively decomposable. The parameter κ is measured in inverse income units, so it has to be chosen for the currency and period at hand.

Theorem 30 (Axiomatic Optimality). *Among the eleven properties of Section 3:*

- (i) *No inequality index satisfies all eleven.*
- (ii) *GE_α with $\alpha < 2$ satisfies every property except translation invariance.*
- (iii) *E_κ with $\kappa > 0$ satisfies every property except scale invariance.*

Proof. Part (i) is Theorem 23. Part (ii) collects Propositions 13–21 for GE_α : it is anonymous, normalized, continuous, population-invariant, and scale-invariant; it satisfies the transfer principle and hence Lorenz consistency (Theorem 26); it satisfies DTS for $\alpha < 2$ (Proposition 18); and it is additively decomposable and therefore subgroup-consistent.

Part (iii). Anonymity, continuity, and population invariance are immediate from (33), and translation invariance holds because $\mu - y_i$ is unchanged when every income rises by c . Normalization: by Jensen's inequality $\frac{1}{n} \sum_i e^{\kappa(\mu - y_i)} \geq e^{\kappa \cdot 0} = 1$, with equality if and only if all incomes are equal, since $x \mapsto e^x$ is strictly convex. Transfer principle: $E_\kappa + 1$ is a sum of the strictly convex function $\phi(y) = e^{\kappa(\mu - y)}$, and a progressive transfer leaves μ unchanged, so the index strictly falls; Lorenz consistency follows by Theorem 26. Diminishing transfer sensitivity: a transfer of δ from $y + d$ to y changes the index by

$$\frac{1}{n} [\phi(y + \delta) + \phi(y + d - \delta) - \phi(y) - \phi(y + d)] = \frac{1}{n} e^{\kappa(\mu - y)} [e^{-\kappa\delta} + e^{-\kappa(d - \delta)} - 1 - e^{-\kappa d}].$$

The bracket is negative and depends only on δ and d , so the reduction is proportional to $e^{-\kappa y}$ and strictly larger at lower y . Additive decomposability: for subgroups k with sizes n_k

and means μ_k ,

$$E_\kappa(\mathbf{y}) + 1 = \frac{1}{n} \sum_k \sum_{i \in k} e^{\kappa(\mu - \mu_k)} e^{\kappa(\mu_k - y_i)} = \sum_k w_k [E_\kappa(\mathbf{y}_k) + 1], \quad w_k = \frac{n_k}{n} e^{\kappa(\mu - \mu_k)},$$

and $\sum_k w_k = E_\kappa(\mu_1 \mathbf{1}_{n_1}, \dots, \mu_K \mathbf{1}_{n_K}) + 1$, which gives (8) with positive weights depending only on subgroup sizes and means. Subgroup consistency then follows from Proposition 27(iii).

E_κ is not scale-invariant: $E_\kappa(1, 3) \neq E_\kappa(2, 6)$ for every $\kappa > 0$. $\square$

Intuition. The variance, the best-known decomposable absolute index, treats transfers at all income levels alike, which suggests that absolute measures cannot be bottom-sensitive. That is a property of the variance, not of translation invariance. The absolute decomposable indices are exactly the variance and the exponential family (Chakravarty and Tyagarupananda, 1998; Bosmans and Cowell, 2010), just as the relative decomposable indices are the GE family, and in each class the members with the right curvature are bottom-sensitive. The choice between relative and absolute measurement is therefore a choice about which invariance one believes, not a choice between a better and a worse axiomatic package. Section 7 shows that it is also the choice with the largest consequences in practice.

7 Do the Axioms Bind? Evidence from Survey Data

The previous sections say which indices satisfy which properties. They do not say how often the differences matter. Two relative indices that both satisfy the transfer principle must agree whenever one Lorenz curve dominates another, and can disagree only when the curves cross. How often curves cross, and how often the indices then disagree, is an empirical question. Atkinson (1970) posed it for 12 countries and Figini (2000) for 69 distributions from the Luxembourg Income Study; we answer it for national income and consumption distributions from 143 countries over six decades, and we add the comparison between relative and absolute indices.

7.1 Data and Design

We use the percentile file of the World Bank’s Poverty and Inequality Platform ([World Bank, 2026a,b](#)). For each country and survey year it reports the mean income or consumption of each of 100 equal-population groups, in 2017 PPP dollars per day. The file covers 2,588 distributions from 172 countries between 1963 and 2025. Most are computed from household microdata; the rest are reconstructed by the platform from grouped data or, for a few countries such as China, synthesised from separate urban and rural distributions. We treat each distribution as a population of 100 equally weighted incomes and compute every index of Section 4 on it, with the 90/10 ratio defined as in (17), the ratio of the 90th to the 10th percentile group mean. We also compute three absolute indices: the standard deviation, the absolute Gini μG (half the mean absolute difference), and the Kolm index with $\kappa = 0.1$ per dollar-day. The theorems apply to any finite population, so they apply exactly to these 100-point distributions. Grouping into percentiles removes inequality within each percentile, which lowers every index slightly. The loss is small: our Gini differs from the platform’s official Gini, computed from the underlying data, by at most 0.0039, and our MLD differs from the official MLD by at most 0.017.

The platform groups surveys into *comparable spells*, within which it judges the welfare concept, survey design, and coverage to be comparable. We compare each national survey with the next survey in the same spell, which yields 1,955 pairs from 143 countries. For each pair we record the sign of the change in every index and the relationship between the two Lorenz curves at the 99 interior percentiles: dominance (no crossing), a single crossing, or multiple crossings. Several indices rank distributions identically—the CV and GE_2 , A_1 and the MLD, A_2 and GE_{-1} , $A_{0.5}$ and $GE_{0.5}$ —so when we ask whether “the indices” agree we use one index per ordering: the Gini, Z^* , GE_{-1} , the MLD, $GE_{0.5}$, the Theil index, and GE_2 . These seven satisfy the transfer principle; the four GE indices with $\alpha < 2$ also satisfy diminishing transfer sensitivity. The platform does not publish standard errors, so we cannot test whether a small change is statistically meaningful; instead we report results by the size

of the change and with a noise screen described below.

7.2 What the Theorems Predict

When one Lorenz curve dominates, Theorem 26 requires every scale-invariant index that satisfies the transfer principle to move in the same direction (scale invariance lets the theorem compare distributions with different means). Dominance holds in 38% of pairs (738 pairs), and in every one of them all seven indices move in the direction the dominance implies. The two indices that violate the transfer principle behave differently. The 90/10 ratio moves against the Lorenz ranking in 56 of the 738 dominance pairs. The variance of logarithms does so in 0: its violations of the transfer principle arise from transfers among incomes well above the geometric mean, as in the counterexample of Proposition 17, and in these data such changes never outweighed the rest of the change in the distribution.

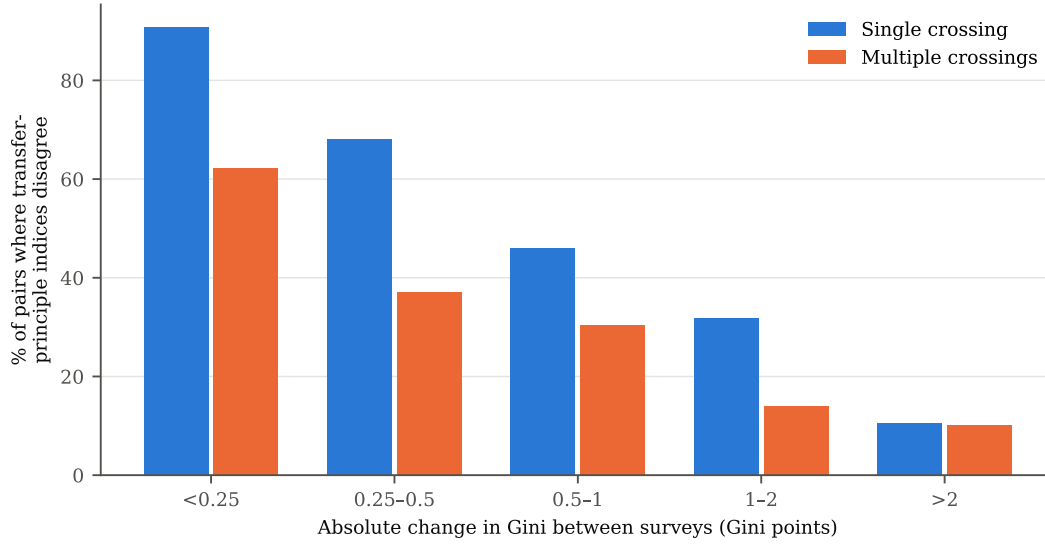
When the curves cross exactly once, Shorrocks and Foster (1987) show that every relative index satisfying the transfer principle and their notion of transfer sensitivity ranks the distribution whose curve is higher at the bottom as more equal, provided that distribution also has the lower coefficient of variation. Their transfer sensitivity is defined through composite, variance-preserving transfers rather than our pairwise Property 9, but GE_α with $\alpha < 2$ satisfies both. The condition holds in 116 single-crossing pairs, and the four transfer-sensitive indices contradict the prediction in 0 of them. GE_2 agrees by construction, since the condition selects on the CV; the Gini and Z^* , which carry no such guarantee, also never contradict the prediction, so these pairs do not discriminate among indices.

7.3 How Often Relative Indices Disagree

Lorenz curves cross in 62% of pairs: 38% cross once and 24% cross more than once. This is in line with earlier evidence. Atkinson (1970) found crossings in 50 of 66 pairwise comparisons of 12 countries, and Figini (2000) notes that the share depends on how finely the curves are compared, from 25 percent with deciles (Bishop et al., 1991) to 80 percent with 100 points

(Buhmann et al., 1988). Using only the nine decile ordinates, our curves cross in 36% of pairs, because many crossings occur in the tails, where the curves are close. Dominance at the decile level does not, however, guarantee that the indices agree; the theorems require dominance at every point of the distribution being measured.

When the curves cross, the seven indices agree on the direction of change in only 51% of pairs, and the four transfer-sensitive indices agree in 65%. Most of this disagreement concerns small changes. Figure 1 sorts pairs by the absolute change in the Gini. When the Gini moves by less than a quarter of a point, the indices disagree in most single-crossing pairs; when it moves by more than two points, they disagree in about one pair in ten. Among the 296 crossing pairs in which the Gini moves by at least one point, they disagree in 23%. Because the Gini is itself one of the indices, we also use a screen that treats the indices symmetrically: in each pair we compare the signs only of those indices whose log change exceeds that index’s own median absolute log change across all pairs. Among the 580 crossing pairs with at least two such indices, they agree in 87%. Without standard errors we cannot say how many of the remaining disagreements reflect sampling noise. The results are the same if we drop the distributions that are not computed from microdata: in the 1,874 pairs of microdata surveys, the curves cross in 63% of pairs and the seven indices agree in 51% of crossing pairs.

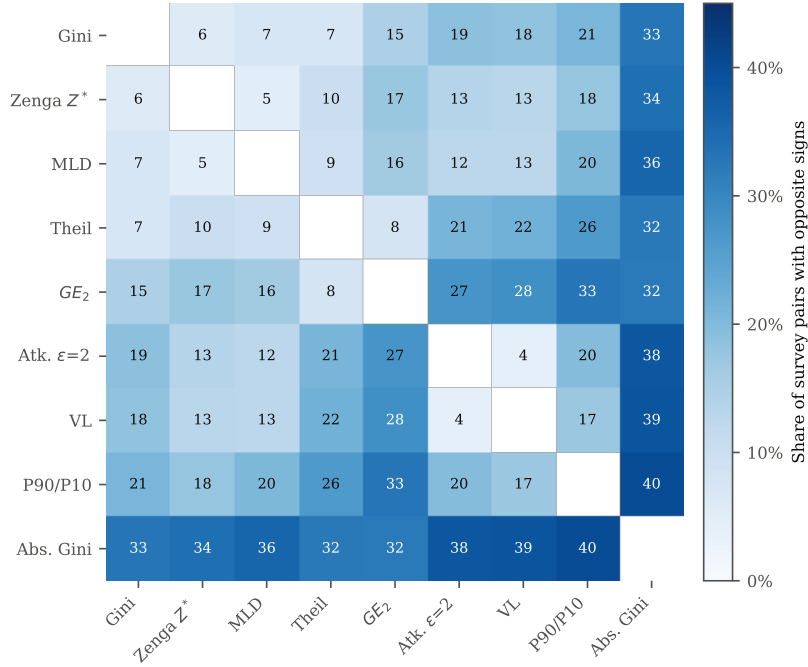


Lorenz-dominance pairs: 0% in every bin, as the transfer principle requires.

Transfer-principle indices (one per distinct ordering): Gini, Zenga Z^* , GE_{-1} , MLD, $GE_{0.5}$, Theil, GE_2 .

Figure 1: Disagreement among transfer-principle indices, by size of change
Notes: Share of consecutive comparable PIP survey pairs in which the seven transfer-principle indices do not all move in the same direction, by Lorenz relationship and by the absolute change in the Gini. Under Lorenz dominance the share is zero in every bin, as Theorem 26 requires.

Which indices disagree is predictable from their sensitivities (Figure 2). Indices with similar weighting rarely conflict: the Gini and the MLD move in opposite directions in 7% of pairs, and the Gini and Z^* in 6%. Indices that weight opposite ends of the distribution conflict often: GE_2 , which is sensitive to the top, and the Atkinson index with $\varepsilon = 2$, which is sensitive to the bottom, disagree in 27% of pairs.



Cells: % of 1,955 consecutive comparable PIP survey pairs in which the two indices move in opposite directions.

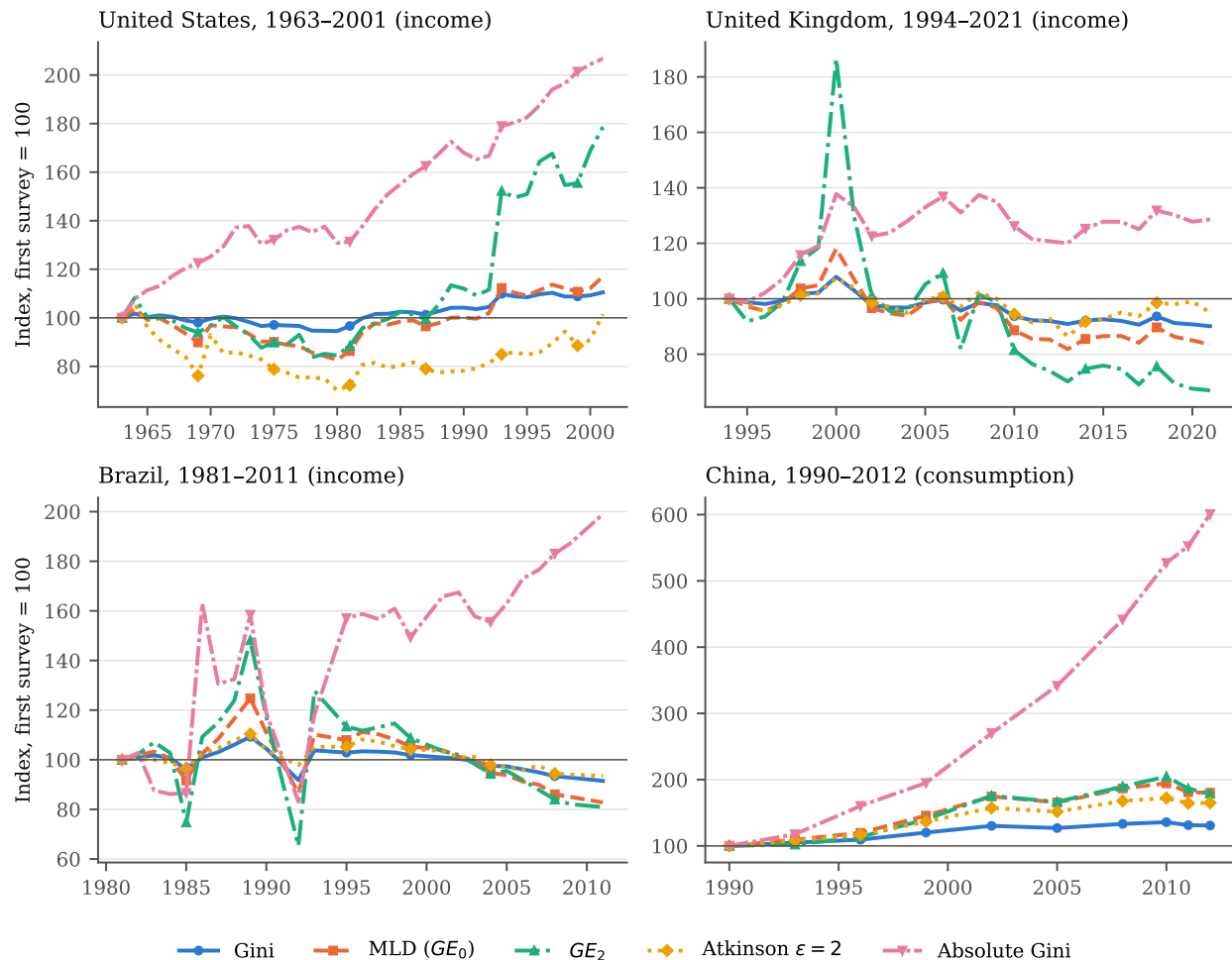
Figure 2: Pairwise disagreement on the direction of change
Notes: Each cell is the percentage of the 1,955 survey pairs in which the two indices change in opposite directions.

7.4 Relative Versus Absolute

The Gini and the absolute Gini move in opposite directions in 33% of pairs, a rate matched among relative indices only by pairs that weight opposite tails, such as GE_2 and the 90/10 ratio (33%). The conflict has a simple arithmetic source. Since $\Delta \ln(\mu G) = \Delta \ln \mu + \Delta \ln G$, the absolute Gini rises whenever growth in the mean exceeds the proportional fall in the Gini. In the PIP pairs the median absolute change in log mean income is 3.8 percent, against 2.1 percent for the log Gini, and growth is the larger of the two in 69% of pairs. Hence, in 81% of the conflicts relative inequality falls while absolute inequality rises. Among the 1,384 pairs in which mean income rose, the Gini fell in 53%, and in 71% of those the absolute Gini rose. Over the 136 comparable spells that span at least ten years, the Gini fell with rising mean income in 77 spells, and the absolute Gini rose in 86% of them. The choice of absolute index matters little: the Kolm index moves in the same direction as the absolute Gini in 96% of

pairs and the standard deviation in 89%, although the Kolm index depends on the unit of κ , which has to be fixed across countries whose mean incomes differ by a factor of more than 50.

The consequence is that Lorenz dominance, which settles the comparison for every relative index satisfying the transfer principle, does not settle the absolute one. Among the 458 pairs in which the Gini moves by at least one point and one Lorenz curve dominates, the seven relative indices agree by Theorem 26, but the absolute Gini moves the other way in 20%. In the 296 crossing pairs with changes of the same size, the relative and absolute Gini conflict in 26%, about as often as the relative indices conflict among themselves (23%). Taking all 754 large changes together, the relative and absolute Gini conflict in 23% of pairs and the relative indices in 9%. Figure 3 shows four cases. In China between 1990–2012, the Gini of consumption rose from 0.322 to 0.421, while mean consumption grew by a factor of 4.6 and the absolute Gini by a factor of 6.0. In Brazil between 2001–2011, the last decade of the spell in the figure, the Gini fell from 0.584 to 0.529, yet the absolute Gini rose by a factor of 1.2 because mean income grew faster than relative inequality fell. In the United States and the United Kingdom the Gini ends within about ten percent of its starting value while the absolute Gini rises by up to a factor of two in the United States. The same disagreement drives the debate over global inequality (Ravallion, 2004; Atkinson and Brandolini, 2010; Niño Zarazúa et al., 2017); the PIP data quantify it for national trends in 143 countries.



Source: World Bank PIP 100-bin distributions (2017 PPP); one PIP comparable spell per panel.

Figure 3: Relative and absolute indices over time in four countries

Notes: Each index is set to 100 in the first survey of one long comparable PIP spell per country. The absolute Gini is μG in 2017 PPP dollars per day. China’s national distributions are synthesised by the platform from urban and rural surveys.

8 Practical Guidance for Applied Researchers

The results of the previous sections reveal a landscape of tradeoffs. No single index satisfies all desirable properties, and the choice of measure should be guided by the research question. We offer the following practical recommendations.

If decomposability matters, use the GE family. For any study that requires decomposing inequality into within-group and between-group components—for example, de-

composing wage inequality by education, gender, or race—the Generalized Entropy family is the only scale-invariant option (Theorem 24). Within the GE family, the choice of α should reflect the researcher’s judgment about which part of the distribution matters most:

- $\alpha = 0$ (MLD): more sensitive to changes at the bottom than the Theil ($\alpha < 0$ is more so). Suitable for poverty-focused analyses.
- $\alpha = 1$ (Theil): balanced sensitivity; weights by income shares in decomposition. Suitable for general-purpose analysis.
- $\alpha = 2$ (half-squared CV): more sensitive to changes at the top ($\alpha > 2$ is more so). Suitable for analyses focused on top-income dynamics.

If welfare-theoretic foundations matter, use the Atkinson index. The Atkinson index has a direct welfare interpretation through the equally distributed equivalent income. It satisfies all properties except translation invariance and additive decomposability, and it satisfies subgroup consistency, which is often sufficient for practical purposes. The parameter ε has a clear interpretation as inequality aversion, making results transparent and facilitating sensitivity analysis.

If simplicity and communication matter, the Gini is defensible—with caveats. The Gini coefficient is intuitive, bounded between 0 and 1, and has a geometric interpretation via the Lorenz curve. It satisfies the transfer principle and Lorenz consistency. However, researchers should be aware of its limitations: it cannot be decomposed additively, it fails subgroup consistency, and its transfer sensitivity depends on ranks rather than income levels. It is a reasonable summary statistic but a poor tool for decomposition.

Prefer the MLD to the variance of logarithms. The VL fails the Pigou-Dalton transfer principle, so it can rise after a transfer from rich to poor. In the PIP data this failure never reversed a Lorenz ranking (Section 7), so VL trends for national distributions are unlikely to be misleading on this account; the risk is greater for distributions with long upper tails, where transfers among top incomes carry more weight. The MLD (GE_0) is also

built on logarithms, satisfies the transfer principle, and decomposes additively, so it is the natural replacement.

Treat the Zenga index as a descriptive device. The Zenga curve $1 - M^-(p)/M^+(p)$ shows how inequality varies across quantiles, and Z^* is population-invariant and satisfies the transfer principle. But it fails diminishing transfer sensitivity and subgroup consistency (Propositions 18 and 20), so it offers no axiomatic advantage over the Gini, and the split-point version Z should be avoided when populations differ in size (Theorem 28).

Percentile ratios are complements, not substitutes. The 90/10 ratio and similar measures are useful descriptive statistics that are robust to outliers and easy to interpret. However, they fail most axiomatic properties and should not be used as the sole measure of inequality. They are best used alongside a full-distribution measure like a GE index or the Gini.

Check the Lorenz curves first, and state the invariance axiom. If one Lorenz curve dominates the other, every relative index satisfying the transfer principle gives the same answer and the choice among them is immaterial. If the curves cross, report indices that weight the bottom and the top differently, such as the Atkinson index with $\varepsilon = 2$ and GE_2 , which disagree in 27% of the PIP comparisons. Above all, say whether the conclusion is about relative or absolute inequality: even when Lorenz dominance makes every transfer-principle relative index agree on a change of a Gini point or more, the absolute Gini disagrees in 20% of the PIP comparisons (Section 7).

9 Conclusion

We evaluated eight inequality indices against eleven properties and then asked how often the properties decide anything in survey data.

On the theory side, three points bear repeating. The maximum number of properties one index can satisfy is ten, and a relative index (GE_α , $\alpha < 2$) and an absolute index (E_κ)

both reach it, so the relative–absolute choice cannot be settled by counting axioms (Theorem 30). The GE family remains the only scale-invariant family that decomposes additively (Theorem 24). The Zenga indices Z and Z^* satisfy the transfer principle (Zenga’s own finite formula Z_N does not), but none of the three satisfies diminishing transfer sensitivity or subgroup consistency, and its split-point form cannot be population-invariant (Theorem 28).

On the empirical side, the transfer principle and the Lorenz ordering it implies settle the direction of change in relative inequality in only 38% of consecutive PIP survey pairs. In the remaining pairs, relative indices that all satisfy the transfer principle often disagree, mostly about small changes; once the Gini moves by a point or more, they disagree in 23% of crossing pairs. The variance of logarithms, whose failure of the transfer principle is a textbook warning, never contradicted a Lorenz ranking in our 738 dominance pairs. The relative–absolute split is different in kind: it persists where the transfer principle settles every relative comparison, with the absolute Gini moving against a Lorenz-dominance ranking in 20% of the large changes in which one Lorenz curve dominates. A researcher who reports whether inequality rose should therefore say which invariance axiom the answer rests on.

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